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Heterogeneity Evaluation Parameter

1. Calculation Average Permeability α_k, β_k and v_k

- $k_{\max} = 435\text{md}$
- $k_{\min} = 9\text{md}$

The average permeability is then given by:

$$\bar{k} = \frac{\sum_{i=1}^n k_i h_i}{\sum_{i=1}^n h_i} = 101.267\text{md}$$

Permeability Contrast is given by:

$$\alpha_k = \frac{k_{\max}}{k_{\min}} = \frac{435}{9} = 48.333$$

The Homogeneity Coefficient is:

$$\beta_k = \frac{\bar{k}}{k_{\max}} = \frac{101.2678571}{435} = 0.23$$

The Coefficient of permeability variation is:

$$v_k = \frac{1}{\bar{k}} \sqrt{\frac{1}{n} \sum_{i=1}^n (k_i - \bar{k})^2} = 0.875$$

Since $\alpha_k = 48.333 > 6$ which is an extremely high permeability contrast thus strong heterogeneity.

Also, $\beta_k = 0.233 < 0.5$ indicates a very low homogeneity coefficient thus strong heterogeneity.

Finally, $V_k = 0.875 > 0.6$, high variation coefficient thus strong heterogeneity.

2. Calculation of D-P variation

No	K-md	h-ft	with greater k	% of h with greater k	$(k_i - \bar{k})^2$
1	435	2	0	0.000	11377.143
2	254	2	2	3.571	3327.1075
3	196	2	4	7.143	974.17889
4	172	3	6	10.714	003.03603
5	166	2	9	16.071	190.25032
6	158	2	11	19.643	218.53603
7	153	2	13	23.214	2676.2146
8	147	2	15	26.786	091.42889
9	128	1	17	30.357	14.607462
10	105	2	18	32.143	3.9288903
11	100	1	20	35.714	.60746173
12	91	2.5	21	37.500	105.42889
13	90	2.5	23.5	41.964	26.964605
14	83	2	26	46.429	33.714605
15	72	1.5	28	50.000	56.607462
16	62	2	29.5	52.679	1541.9646
17	49	2.5	31.5	56.250	731.92889
18	49	2	34	60.714	731.92889
19	49	2	36	64.286	731.92889
20	44	2	38	67.857	279.60746
21	40	2	40	71.429	753.75032
22	35	2	42	75.000	391.42889
23	32	2	44	78.571	798.03603
24	30	2	46	82.143	079.10746
25	28	2	48	85.714	368.17889
26	27	2	50	89.286	5515.7146
27	20	2	52	92.857	6604.4646
28	9	2	54	96.429	513.35746
	Total	56	758.5		

Table 1: Calculation of DP variation V

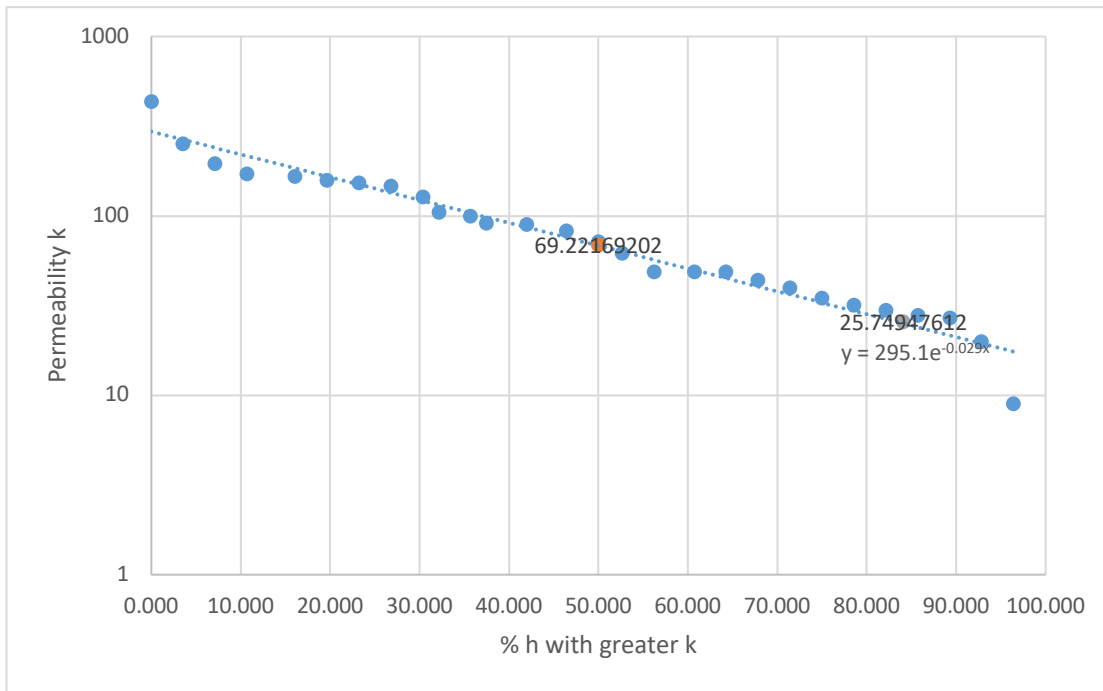


Fig 1: % of h with greater k vs Permeability

$$V = \frac{k_{50} - k_{84.1}}{k_{50}} = \frac{69.22 - 25.75}{69.22} = 0.63$$

Interpretation of Results

Conclusion for Dykstra-Parsons Coefficient $V = 0.63$.

This value falls in the upper range of moderate heterogeneity, approaching strong heterogeneity.

Taken together, these metrics demonstrate that the reservoir is strongly heterogeneous. The extreme differences in permeability, combined with high overall variability, result in an uneven distribution of fluid flow. This leads to inefficient sweep efficiency, meaning that some areas of the reservoir will be well-flushed while others remain largely bypassed, reducing overall recovery performance.

3. Lorenz L Coefficient

	h-md	-ft	n h	kh	m kh	normalized sum h	normalized sum kh	trapezoid_area
						0	0	0
1	435	2	2	870	870	0.035714286	0.153412097	0.002739502
2	254	2	4	508	378	0.071428571	0.242990654	0.004798851
3	196	2	6	392	270	0.107142857	0.312114266	0.006171751
4	172	3	9	516	286	0.160714286	0.403103509	0.012186059
5	166	2	11	332	2618	0.196428571	0.461646976	0.01045419
6	158	2	13	316	2934	0.232142857	0.517369071	0.011940449
7	153	2	15	306	2240	0.267857143	0.571327808	0.013489684
8	147	2	17	294	2534	0.303571429	0.623170517	0.014812202
9	128	1	18	128	2662	0.321428571	0.645741492	0.00705343
10	105	2	20	210	2872	0.357142857	0.682771998	0.012563922
11	100	1	21	100	2972	0.375	0.700405572	0.006455148
12	91	1.5	22.5	27.5	299.5	0.419642857	0.740521954	0.015939098
13	90	1.5	26	225	2424.5	0.464285714	0.780197496	0.017535173
14	83	2	28	166	2590.5	0.5	0.809469229	0.014113157
15	72	1.5	29.5	108	2598.5	0.526785714	0.82851349	0.009777187
16	62	2	30.5	124	2322.5	0.5625	0.850379122	0.01190896
17	49	1.5	34	22.5	2945	0.607142857	0.87198025	0.012632803
18	49	2	36	98	25043	0.642857143	0.889261153	0.010800564
19	49	2	38	98	25141	0.678571429	0.906542056	0.011417739
20	44	2	40	88	25229	0.714285714	0.922059601	0.010806862
21	40	2	42	80	25309	0.75	0.936166461	0.010328236
22	35	2	44	70	25379	0.785714286	0.948509963	0.009478046
23	32	2	46	64	25443	0.821428571	0.959795451	0.009068695
24	30	2	48	60	25503	0.857142857	0.970375595	0.008879764
25	28	2	50	56	25559	0.892857143	0.980250397	0.008640451
26	27	2	52	54	25613	0.928571429	0.989772527	0.00867194
27	20	2	54	40	25653	0.964285714	0.996825957	0.006675567
28	9	2	56	18	25671	1	1	0.003117364
total	2824	56		671				282456798
							Lorenz L	0.4350864

Table 2: Calculation of Lorenz L Coefficient

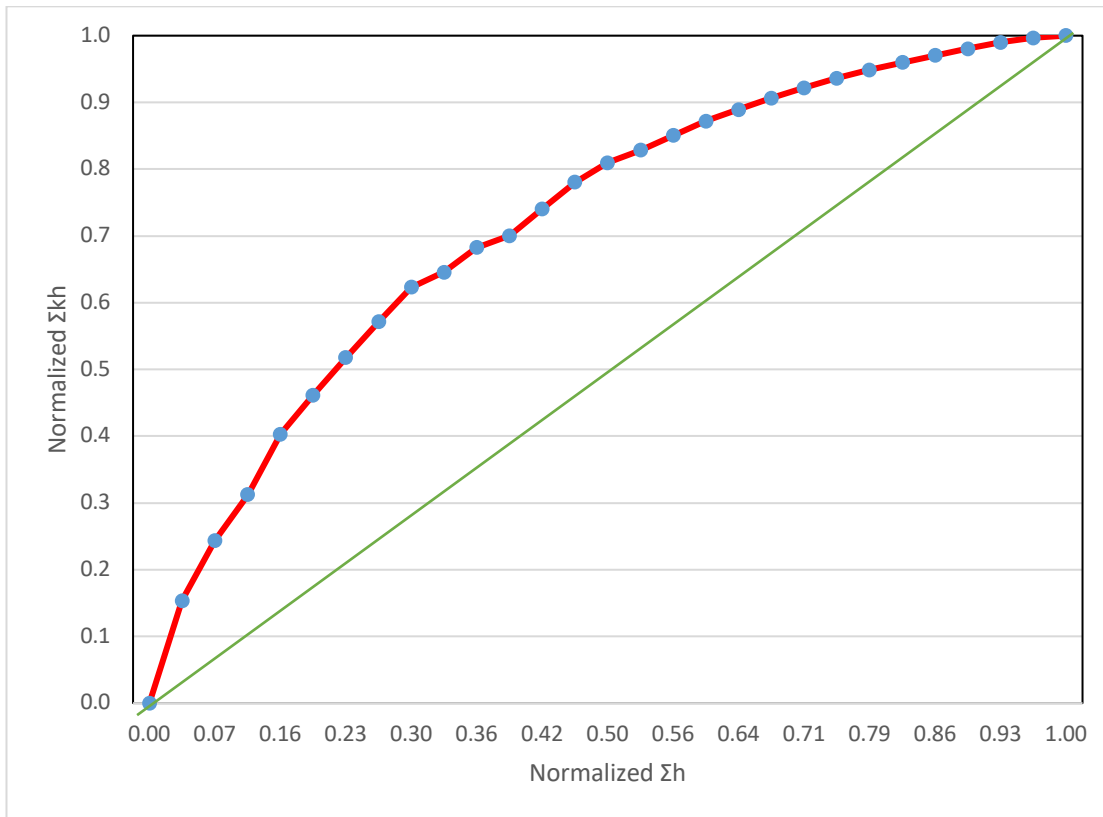


Fig 2: Normalized Σh vs Normalized Σkh

$$\text{Lorenz Coefficient } L = \frac{\text{Area above the straight line}}{\text{Area below the straight line}} = \frac{0.5 - 0.28}{0.5} = 0.44.$$

- Interpretation of Results

heterogeneous (Dykstra-Parsons $V = 0.63$; Lorenz $L = 0.44$), showing significant permeability variation that causes partially uneven fluid flow and requires targeted management to optimize recovery.